# Experimental study of acoustic resonance in a rectangular cavity

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The resonance frequency and sound pressure spatial distribution within a cavity offer highly interesting applications in acoustics, such as the acoustic design of buildings and the design and tuning of musical instruments. This experiment studies whether the measured resonance frequencies within a rectangular wooden cavity match theoretical predictions derived from solving the three-dimensional wave equation with ideal rigid boundary conditions. A Bluetooth speaker and a microphone were used to measure sound pressure amplitude as a function of driven frequency inside the cavity. Six significant resonance frequencies were identified within the $0 - 800\ Hz$ range at $147 \pm 3.5\ Hz$, $308 \pm 4.5\ Hz$, $444 \pm 4.5\ Hz$, $588 \pm 4.5\ Hz$, $735 \pm 4.0\ Hz$, $799 \pm 5.0\ Hz$. These measurements were compared to theoretical predictions corresponding to the most likely resonance modes using a statistical t-score analysis. The results showed good overall agreement, with five modes having t-scores below $0.76\sigma$ and only one mode showing slight tension ($1.67\sigma$ difference). However, the statistical t-score analysis showed no significant difference between the predicted resonance frequencies and the measured resonance frequencies; a systematic underestimation of the measured values by the predicted values is evident in the normal distribution of the frequency values. This discrepancy is mainly caused by non-ideal boundary behavior, such as wall vibrations that result in an entire shift in the sound pressure spatial distribution. These results show that while the idealized model used can effectively predict the natural resonance frequencies inside the enclosed rectangular cavity, it is important to consider boundary effects to improve the accuracy of theoretical predictions.

# I. INTRODUCTION

This experiment explores acoustic resonance in a cavity, and a relevant application is understanding how sound is amplified or diminished in enclosed building spaces. Intuitively, we might expect that in an enclosed environment with a fixed sound source, the sound volume decreases with distance, but due to resonance effects, the actual distribution of sound pressure can be far more complex. Several key concepts are central to this experiment:

- **Standing wave**: Two waves traveling in opposite directions with the same amplitude then interfere and create a standing wave [1].
- **Cavity natural resonance frequencies**: Resonance occurs when a given system is driven externally to vibrate at greater amplitude at certain preferred frequencies. These are known as resonance frequencies, which correspond to the natural vibration frequencies of the object. For an acoustical cavity, which is usually a multi-mode resonant system, there are multiple resonance frequencies. These resonance frequencies will depend on the dimensions of the cavity, its geometry, and the speed of sound at the temperature of the air [2].
- **Sound pressure**: The change in pressure caused by vibration when sound waves pass through the air, and the air is the medium in this experiment.

Additionally, the theoretical framework relies on solving the three-dimensional wave equation by using the separation of variables and applying the ideal gas law to determine the speed of sound. Since air can be approximated as an ideal gas, the speed of sound in air at temperature $T$ can be calculated using the following formula:

- $v_{ideal} = \sqrt{\frac{\gamma k T}{m}}$, where $\gamma$ is the adiabatic index, $T$ is the temperature and $k$ is the Boltzmann constant.

In this experiment, we used a rectangular wooden cavity as the object, placed a Bluetooth speaker at the center of the bottom of the cavity, and placed a microphone connected to an oscilloscope at the midpoint of the cavity's bottom edge to analyze the sound pressure distribution. We set the frequency of sound pressure as the independent variable, and measured the corresponding changes in sound pressure amplitude displayed on the oscilloscope as the dependent variable. The measured resonance frequency is compared to the predicted resonance frequency of the possible resonance mode. The agreement between theory and experiment was evaluated by comparing the values in units of standard deviation $\sigma$, using the t-score method. **Research Question:** Do the experimentally measured resonance frequencies align with the theoretical predictions for some resonance modes?

# II. METHODS

## A. Experimental Design

We present the theoretical model that we applied to a rectangular cavity. For an enclosed box cavity, we assumed the boundary conditions for the sound pressure $P$ [3]:

$$\left.\frac{\partial P}{\partial x}\right|_{x=0} = \left.\frac{\partial P}{\partial x}\right|_{x=L_x} = \left.\frac{\partial P}{\partial y}\right|_{y=0} = \left.\frac{\partial P}{\partial y}\right|_{y=L_y} = \left.\frac{\partial P}{\partial z}\right|_{z=0} = \left.\frac{\partial P}{\partial z}\right|_{z=L_z} = 0$$

Where $L_x, L_y, L_z$ represent length, width, and height. Then consider solving the wave equation:

$$\nabla^2 P = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + \frac{\partial^2 P}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 P}{\partial t^2},$$

by using the separation of variables, which led to the theoretical resonance frequency model [3]:

$$f_{n_x n_y n_z} = \frac{\omega_{n_x n_y n_z}}{2\pi} = \frac{v}{2}\sqrt{\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}},$$

where $f_{n_x n_y n_z}$ represents the theoretical resonance frequency, and $n_x, n_y, n_z$ are the number of antinodes in the length, width, and height directions of the cavity.

Using a ruler, first we measured the height, width, and length $(L_x, L_y, L_z)$ of the wooden cavity. The uncertainties of $L_x$ and $L_y$ were estimated to be $\pm 1\ cm$, mainly due to the ruler we used was not straight and the eye alignment errors were large. The uncertainty of $L_z$ was estimated to be $\pm 5\ cm$, mainly due to the ruler we used was not long enough. For the reader, the uncertainties of geometric dimensions depend on the accuracy of their experimental equipment and they should select appropriate values.

A Bluetooth speaker was placed right in the middle at the bottom to uniformly excite the symmetry of the square x-y plane sound wave, and a microphone was placed close to the midpoint of the bottom edge of the cavity to avoid nodes and avoid the degenerate modes caused by $L_x = L_y$. Then we closed the cavity to approximate the assumed boundary conditions. We then connected the microphone to an external oscilloscope via an amplifier using a cable, and the Bluetooth speaker was used to play the drive signal. Figure 1(a) shows all the equipment required for this experiment and their arrangement in the $z - y$ cross-section. Figure 1(b) provides a 3-D representation of the exact positions of the microphone and speaker within the rectangular cavity.

To identify resonance peaks, we first swept the driving frequency of the speaker from 0 to $800Hz$. Initially, we rough located the possible resonance region in $20Hz$ steps. Once the peak was identified, we determined its exact position near the initial estimate in only $1Hz$ steps. When a significant decrease followed by a increase in amplitude was observed

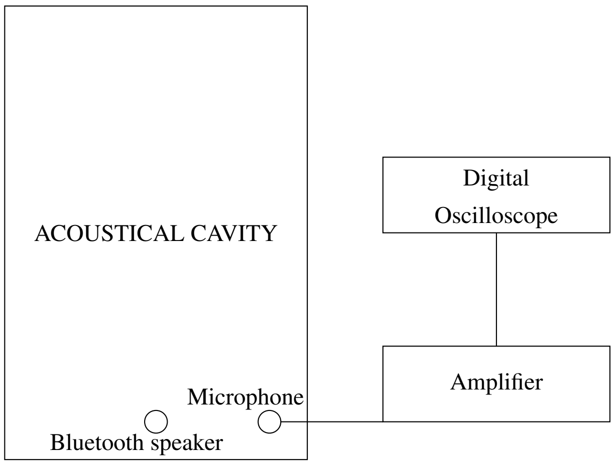


FIG. 1(a): Cross-sectional view of the experimental setup in the z-y plane, showing the connections of the speaker, microphone, amplifier, and the oscilloscope.

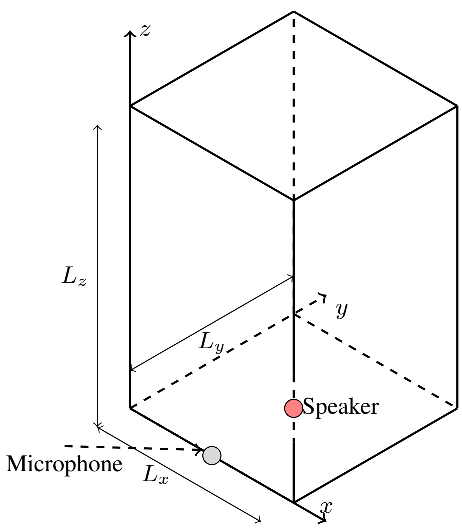


FIG. 1(b): Three-dimensional view of the acoustic cavity, showing the coordinate axes and the locations of the speaker and microphone.

on the oscilloscope, the corresponding frequency was identified as a resonance frequency. To characterize the corresponding mode, we moved the microphone from the bottom to the top of the cavity while maintaining the same driving frequency. By analyzing amplitude variations, we determined the number of anti-nodes in the $z$-direction ($n_z$). Since multiple resonance modes can share the same value of $n_z$, we used it to reduce the number of candidate modes, and selected the $(n_x, n_y, n_z)$ combination whose predicted resonance frequency best matched the measured one. This procedure was repeated for all significant resonance peaks within the $0 - 800Hz$ range.

### B. Uncertainty Analysis

The uncertainties in both the measured and predicted resonance frequencies come from several sources. In this section, we distinguish between uncertainties that are included in the final results estimate and those that are only identified but are not included in the propagated uncertainty.

**Measured frequency uncertainties:** The uncertainties in the measured resonance frequencies were estimated based on the observed width of each resonance peak. More Specifically, for each peak, we identified the frequency interval over which the amplitude remained within approximately $68\%$ of the maximum value, and took "half the width of this interval" as the one sigma uncertainty of that resonance frequency. This approach accounts for multiple sources, such as oscilloscope fluctuations, microphone response, and environmental noise. Therefore, we did not separately add uncertainties from step size or background noise. The resulting uncertainties were used in all t-score comparisons.

**Predicted frequency uncertainties:** The uncertainties in the predicted frequencies were derived through propagation from uncertainties in the cavity's geometric dimensions. We used the general formula of uncertainty propagation [4]:

$$u[R] = \sqrt{\sum_{i=1}^{N} (u[x_i] \frac{\partial R}{\partial x_i})^2},$$

Therefore,

$$uf_{n_x n_y n_z} = \frac{v^2}{4f} \sqrt{\left(\frac{n_x^2}{L_x^3} uL_x\right)^2 + \left(\frac{n_y^2}{L_y^3} uL_y\right)^2 + \left(\frac{n_z^2}{L_z^3} uL_z\right)^2}.$$

In this calculation, the speed of sound, which is approximated according to the ideal gas formula and the value of room temperature, is considered to be constant, and then its uncertainty is ignored.

**Unquantified impact (not included):** Several effects were identified as potential sources of systematic error, but were not included in the numerical uncertainty estimates, for example, the fact that the wooden cavity walls are not ideally rigid and the swinging of the microphone position during the experiment.

### C. Result Test

Both the measured and theoretically predicted resonance frequencies are associated with uncertainties corresponding to a $68\%$ confidence interval (one standard deviation $\sigma$) [5]. To quantitatively evaluate the agreement between measured and predicted resonance frequencies, we considered the t-score method from PHYS 119 [6]. Given two values with uncertainties, $A \pm u[A]$ and $B \pm u[B]$, the t-score is defined as:

$$t = \frac{A - B}{\sqrt{(u[A])^2 + (u[B])^2}}.$$

This score effectively measures how many combined standard deviations (sigmas) separate the two values. Based on the magnitude of $t$, we interpreted the level of agreement as follows:

- $t \leq 1$: The values are statistically consistent (in agreement).
- $1 < t < 3$: The values are in tension; agreement is inconclusive.
- $t \geq 3$: The values are likely to be significantly different.

This classification allows for an objective, uncertainty-aware comparison of experimental and theoretical values. In the abstract and figures, t-scores are reported in units of $\sigma$; for example, a t-score of 2 is expressed as a "2$\sigma$ difference".

## III. RESULTS AND DISCUSSION

We recorded all significant resonance frequencies within the 0-800 Hz range and identified the corresponding most likely resonance mode $(n_x, n_y, n_z)$ for each peak and calculated the theoretical resonance frequency. These results are summarized in Table I.

TABLE I: Comparison of predicted and measured resonance frequencies with uncertainties.

| Resonance Mode | Prediction (Hz) | Measured Frequency (Hz) |
|---|---|---|
| $[0, 0, 1]$ | $143.3 \pm 5.97$ | $147 \pm 3.5$ |
| $[0, 0, 2]$ | $286.7 \pm 11.94$ | $308 \pm 4.5$ |
| $[0, 0, 3]$ | $430.0 \pm 17.92$ | $444 \pm 4.5$ |
| $[0, 0, 4]$ | $573.3 \pm 23.89$ | $588 \pm 4.5$ |
| $[0, 0, 5]$ | $716.7 \pm 29.86$ | $735 \pm 4.0$ |
| $[0, 2, 1]$ | $792.2 \pm 17.39$ | $799 \pm 5.0$ |
| $[2, 0, 1]$ | $792.2 \pm 17.39$ | $799 \pm 5.0$ |

As shown in Table II, we also recorded the geometric properties of the rectangular cavity, the temperature on the day of the experiment, and the ideal speed of sound to make it easy for readers to reproduce our experimental results.

TABLE II: Geometric properties of the cavity and ideal speed of sound. Notice that the uncertainty of temperature and speed of sound is not included, as the speed is estimated from the ideal gas law using approximate room temperature.

| Quantity | Value | Uncertainty |
|---|---|---|
| Length ($L_x$) | $0.44\ m$ | $0.01\ m$ |
| Width ($L_y$) | $0.44\ m$ | $0.01\ m$ |
| Height ($L_z$) | $1.20\ m$ | $0.05\ m$ |
| Room Temperature ($T_0$) | $294.55\ K$ | – |
| Speed of Sound ($v_{\text{ideal}}$) | $344\ m/s$ | – |

As shown in Table I, there is a general agreement between the measured resonance frequencies and the theoretical predictions (notice that resonance modes $[2, 0, 1]$ and $[0, 2, 1]$ are degenerate). To further quantify the level of agreement, we calculated the t-score for each resonance mode based on the measured and predicted values. The results are presented in Table III. To identify the number of antinodes along the $z$-axis ($n_z$), we observed periodic variations in the amplitude of sound pressure while moving the microphone vertically at a fixed driving frequency. The number of amplitude peaks is equal to the number of antinodes along the $z$-direction.

TABLE III: T-score comparison between measured and predicted resonance frequencies.

| Resonance Mode $(n_x, n_y, n_z)$ | t-score ($\sigma$) | Agreement |
|---|---|---|
| $[0, 0, 1]$ | 0.53 | Consistent |
| $[0, 0, 2]$ | 1.67 | In tension |
| $[0, 0, 3]$ | 0.76 | Consistent |
| $[0, 0, 4]$ | 0.60 | Consistent |
| $[0, 0, 5]$ | 0.61 | Consistent |
| $[0, 2, 1]$ | 0.38 | Consistent |
| $[2, 0, 1]$ | 0.38 | Consistent |

Although mixed mode from horizontal components can cause non-uniform spatial sound pressure distributions, we have confirmed that the vertical sound pressure distribution is still sufficiently regular to allow $n_z$ to be calculated reliably. To ensure that our mode assignments were unambiguous, we compared our measured frequencies to theoretical predictions for other possible modes in the same frequency range. As shown in Table IV, alternative low-order modes such as $[1, 0, 0]$ and $[1, 1, 0]$ lie outside the uncertainty bounds of our measured peaks, confirming that mode misidentification is unlikely.

Another point worth noting is that, in the lower frequency range shown in Table I, the identified resonance modes tend to zero antinodes in the $x$ and $y$-directions ($n_x = n_y = 0$). We believe this is due to the fact: in this lower frequency range, resonance peaks associated with more complex spatial modes may have been obscured or overlapped by stronger, more dominant peaks corresponding to simpler modes. This overlap effect can lead to some modes being underrepresented or entirely missed in the experimental data.

**Discussion:** We analyzed the reasons for the above findings. On the one hand, as can be seen in Figure 2(a) and 2(b), there is a consistent trend in which the predicted frequencies are systematically lower than the measured ones, and this directional deviation cannot be a random deviation expected from Gaussian distribution. This systematic underestimate indicates that, even though the identification of resonance modes (particularly $n_z$) is reliable and the prediction can be consistent with the measured values, the theoretical model does not fully capture the physical behavior of the rectangular cavity. This deviation is probably due to two main

TABLE IV: Theoretical predictions for alternative low-order modes near the measured frequency range

| Alternative Mode | Prediction (Hz) | Uncertainty (Hz) |
|---|---|---|
| $[1, 0, 0]$ | 389.6 | 8.8 |
| $[0, 1, 0]$ | 389.6 | 8.8 |
| $[1, 0, 1]$ | 415.1 | 8.5 |
| $[0, 1, 1]$ | 415.1 | 8.5 |
| $[1, 1, 0]$ | 550.9 | 8.8 |
| $[1, 1, 1]$ | 569.3 | 8.7 |
| $[2, 0, 0]$ | 779.2 | 17.6 |
| $[0, 2, 0]$ | 779.2 | 17.6 |
| $[1, 0, 2]$ | 483.7 | 10.0 |
| $[2, 0, 1]$ | 483.7 | 10.0 |

factors:

(i) Overlapping of resonance peaks at the $0 - 800Hz$ frequency range, which makes it difficult to isolate individual modes.

(ii) The effect of a wooden cavity wall, which may vibrate or leak sound, thus contradicting the assumption of a perfectly rigid boundary. Therefore, we provide here a mathematical proof of why walls that are not absolutely rigid can lead to predicted values that are lower than the actual values: When $\frac{\partial P}{\partial n}\big|_{\text{boundary}} = 0$, we can assume $P(x) = A\cos(kx) + B\sin(kx)$ in one dimension, and then the standing wave satisfies $kL = n\pi \rightarrow f_n = \frac{vn}{2L}$. When $\frac{\partial P}{\partial n}\big|_{\text{boundary}} = -\alpha P \neq 0 \quad (\alpha > 0)$, the resonance condition can becomes approximately $\tan(kL) = \frac{\alpha}{k} > 0$, thus if we compared to the theoretical model, we can see that the actual situation will produce an overall frequencies shift.

## IV. CONCLUSIONS

This study set out to analyze the acoustic resonance behavior of a rectangular wooden cavity by comparing experimentally measured resonance frequencies with theoretically predicted values. This study found that while the theoretical model $f_{n_x n_y n_z} = \frac{v}{2}\sqrt{\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}}$ generally predicts the real natural resonance frequency of the rectangular cavity well, there is still a systematic difference between the predicted and experimental values. The results of this experiment demonstrate that a rectangular wooden cavity with dimensions $0.44 \pm 0.01$ m in length and width, and $1.20 \pm 0.05\ m$ in height, shows six significant cavity's natural resonance frequencies at $147 \pm 3.5\ Hz$, $308 \pm 4.5\ Hz$, $444 \pm 4.5\ Hz$, $588 \pm 4.5\ Hz$, $735 \pm 4.0\ Hz$, and $799 \pm 5.0\ Hz$ when driven by sound in the frequency range of $0 - 800Hz$. With an assumed speed of sound of $344\ m/s$, these frequencies correspond respectively to the theoretical resonance modes $[0, 0, 1]$, $[0, 0, 2]$, $[0, 0, 3]$, $[0, 0, 4]$, $[0, 0, 5]$, and $[0, 2, 1]$ (degenerate with $[2, 0, 1]$), with associated t-scores of $0.53\sigma$, $1.67\sigma$, $0.76\sigma$, $0.60\sigma$, $0.61\sigma$, $0.38\sigma$. The final analysis concluded that we can use theoretical model to reasonably predict the cavity's natural resonance frequencies, but there was a systematic underestimation, which cannot be explained by random fluctuations alone. This difference is mainly due to the non-ideal rigidity of the cavity walls, which was supported both by physical observations (such as visible wall vibrations) and by theoretical analysis showing that the boundary condition $\frac{\partial p}{\partial n} = 0$ is not strictly satisfied in reality. The principal theoretical implication of this study is that in real-world acoustic system, for a closed rectangular cavity, we can use the wave equation to predict the natural resonance frequencies of the cavity. At the same time, if we assume ideal rigid boundary conditions for the real cavity, it may lead to a systematic underestimate but still a valid prediction. The limitations of this study are that the oscilloscope cannot resolve closely spaced or overlapping resonance peaks, which limits the resolution of mode identification in more complex regions. Future research could focus on identifying and visualizing the overall standing wave patterns inside the cavity (improving spatial scanning techniques for full three-dimensional mode reconstruction), as well as on analyzing cavity with partially absorbing of flexible walls.


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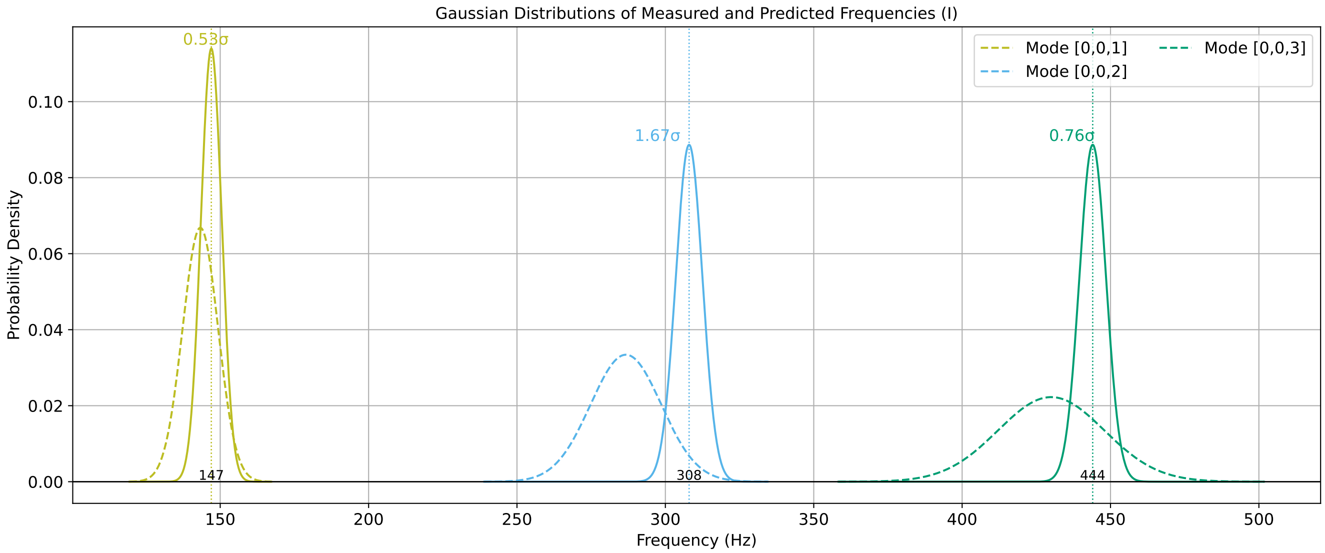


FIG. 2(a): Gaussian distribution of the values of Table 1 for modes $[0,0,1]$, $[0,0,2]$ and $[0,0,3]$, corresponding to measured resonance frequencies of $147\ Hz$, $308\ Hz$ and $444\ Hz$, respectively. The agreement between the predicted and measured values is quantified using the t-score, respectively, showing 0.53 $\sigma$ difference, 1.67 $\sigma$ difference, and 0.76 $\sigma$ difference. At the same time, we will also find in the figure that the predicted center values, which are plotted as dashed lines, are always lower than the measured center values, which are plotted as solid lines.

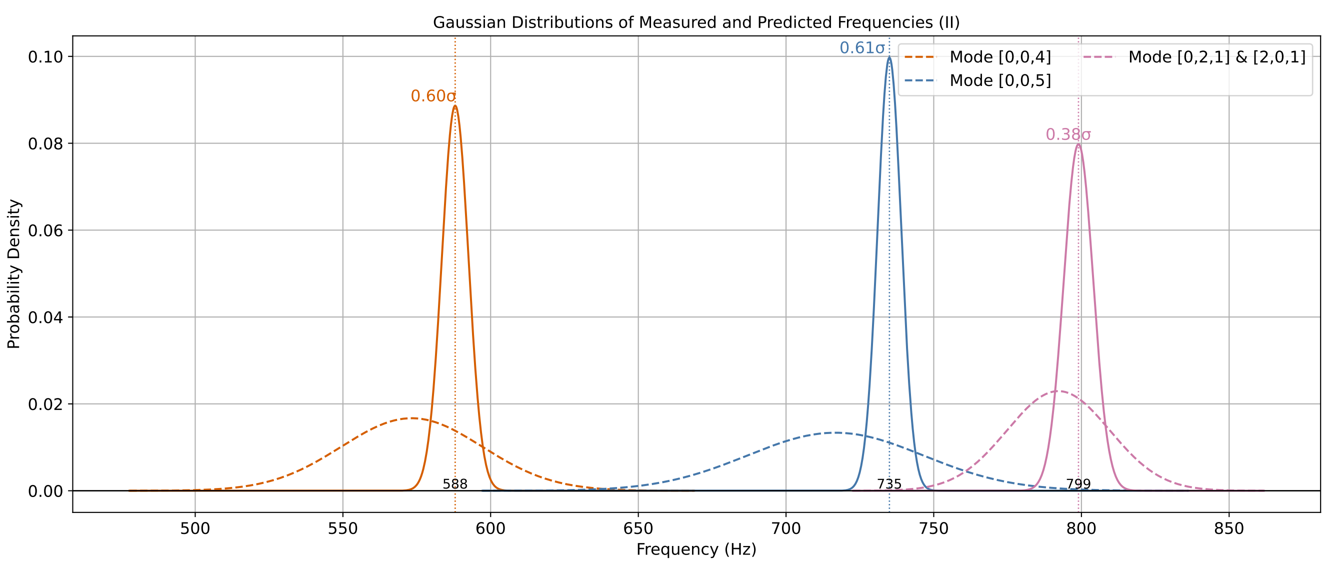


FIG. 2(b): Gaussian distribution of the values of Table 1 for modes $[0,0,4]$, $[0,0,5]$ and the degenerate modes $[0,2,1]$ & $[2,0,1]$, corresponding to measured resonance frequencies of $588\ Hz$, $735\ Hz$ and $799\ Hz$, respectively. The agreement between the predicted and measured values is quantified using the t-score, respectively, showing 0.60 $\sigma$ difference, 0.61 $\sigma$ difference, and 0.38 $\sigma$ difference. At the same time, we will also find in the figure that the predicted center values, which are plotted as dashed lines, are always lower than the measured center values, which are plotted as solid lines.